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Nonsubmodular Cooperation in UAV Swarms for Active Missile Defense

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Defending high-value assets against precision-guided missiles is among the hardest problems in modern autonomous systems, and cooperative swarms of low-cost unmanned aerial vehicles (UAVs) are rapidly changing how this defense can be mounted. We study such a swarm tasked with deploying smoke screen countermeasures, formulating its coordination as a high-dimensional, nonconcave optimal control problem in which the objective of maximizing the union of obscuration intervals is inherently nonadditive. We prove that the underlying discrete selection layer is monotone but *not* submodular, so agent contributions cannot be assessed in isolation and greedy or auction-based allocation carries no near-optimality guarantee, a structural result that motivates joint planning and generalizes beyond the present application. To navigate the resulting 40-dimensional decision space, we propose a joint planning framework driven by an enhanced metaheuristic algorithm (SBOA-Pro). Our simulations demonstrate that optimized cooperation yields a substantial improvement in defensive effectiveness over unoptimized baselines, achieving a total obscuration of 28.73 s. Our analysis identifies the empirical signature of this nonsubmodular structure, a counter-intuitive emergent principle termed Phantom Synergy, in which agents with zero measurable individual contribution can nonetheless be indispensable to the collective solution. This challenges additive agent valuation methods and indicates that in complex cooperative tasks the utility of an agent is defined not by its isolated output but by its role within the system-level solution. A sensitivity analysis further reveals an asymmetric robustness profile. The collective strategy retains roughly 95% of its efficacy under a 6.7% variation in threat velocity, yet remains markedly sensitive to internal execution errors, indicating that its viability depends more on platform execution precision than on external threat uncertainty.

Keywords: Multi-agent systems; optimal control; nonsubmodular optimization; cooperative control; unmanned aerial vehicles.

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1. Introduction

The proliferation of advanced precision-guided munitions challenges traditional defense methods and necessitates agile, intelligent active defense systems [1, 2]. While hard-kill interception remains a staple [3–5], the deployment of multispectral obscurants via Unmanned Aerial Vehicle (UAV) swarms has emerged as a promising soft-kill alternative. Unlike missile-centric defenses that face cost asymmetries and magazine-depth limitations [6], smoke-generating swarms offer a dynamic, reconfigurable barrier capable of scaling against low-cost drone threats and complex sensor environments [7, 8]. The core scientific challenge, therefore, lies in orchestrating these distributed agents to maximize defensive effectiveness.

Cooperative active defense has been approached largely through closed-loop differential games [9, 10] and, more recently, mean-field formulations for large swarms [11, 12], while UAV collectives are coordinated through consensus- and auction-based task allocation [13, 14]. What they share is an objective that decomposes into additive, per-agent or per-engagement terms. The coordinated deployment of multispectral obscurants studied here is structured differently, its collective effect defined by the geometric *union* of occlusion intervals rather than a sum of per-agent rewards.

The difficulty of this problem is structural. It resides in the objective, while the agent dynamics remain simple. We study it as a deterministic open-loop planning problem,

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a restricted static special case of the closed-loop games described above, in which threats and agents follow simple deterministic kinematics. Holding the dynamics fixed isolates the combinatorial structure of the objective, which is where both the hardness and our contribution reside. The effectiveness of a smoke screen is the total time during which *every* critical line of sight to the target is simultaneously blocked, a conjunction of coverage conditions integrated over time, so the objective is nonadditive and the marginal value of one obscurant cloud depends entirely on what the others are doing. The problem has two coupled layers. At the discrete layer of *which* agents and munitions to activate, the induced set function is monotone but *nonsubmodular*. We exhibit an explicit complementarity instance in which activating one obscurant *raises* the marginal contribution of another (Proposition 3.4), so greedy selection carries no standard $(1 - 1/e)$ approximation guarantee. At the continuous layer of placement and timing, the objective is nonconcave and nonsmooth, a time integral of a binary occlusion indicator that is piecewise-constant with gradient zero almost everywhere and available only through a black-box simulator, over a 40-dimensional space that couples the flight and multi-munition release timings of five UAVs. These properties eliminate the natural candidates layer by layer: convex and gradient-based methods (nonconcave, nonsmooth) [15, 16], greedy submodular maximization (nonsubmodular), and exhaustive or grid search (the curse of dimensionality [17, 18]), leaving a pragmatic choice among derivative-free global-search methods. A discrete restricted variant of the deployment problem is NP-hard, but our contribution rests on this structural characterization rather than on worst-case hardness.

Building on this analysis, we develop a systematic framework for planning and analyzing cooperative obscuration strategies. We adopt a derivative-free optimizer (SBOA-Pro) within a progressive analytical protocol that scales from single-agent baselines to full-scale swarm coordination, isolating and quantifying the value of cooperation at each stage. Our contributions are fourfold. At the level of problem structure, we formalize multi-UAV, multi-obscurant defense against multiple lines of sight as the maximization of a nonadditive union-of-occlusion objective, and characterize its two-layer structure of nonsubmodular discrete selection together with nonconcave, nonsmooth, high-dimensional continuous placement. This characterization is what dictates a derivative-free global method (Sec. 3). Methodologically, a staged protocol that progresses from single-agent 4D through temporal and spatial 12D to the full 40D problem, together with a tactically motivated dimensionality reduction, isolates and quantifies cooperative gains that a monolithic solve would obscure (Secs. 3 and 4). Analytically, we derive a closed-form expression for the single-cloud obscuration window whose predictions agree with simulation (Sec. 3.3), which distinguishes this study from a purely numerical solution; the resulting full-scale solution is moreover reproducible, with source code and result data released for independent verification (Sec. 4). Finally, a counterfactual ablation reveals Phantom Synergy, the empirical signature of the nonsubmodular structure, in which an agent whose measured individual utility is zero can still be essential to the collective outcome (Sec. 5).

This paper is structured as follows. Section 2 reviews related work. Section 3 details our methodology, including the system model and the progressive analytical approach. Section 4 presents the results of our analysis. Section 5 provides an in-depth discussion of the emergent tactics and the Phantom Synergy principle. Finally, Sec. 6 concludes the paper. The overall research roadmap is visually summarized in Fig. 1.

2. Related Works

2.1. *Cooperative defense and task allocation*

Active missile defense has been studied most rigorously through differential game theory [9, 10]. The canonical target-attacker-defender formulation derives closed-loop interception guidance under adversarial engagement dynamics [19–21], with extensions to multiplayer pursuit-evasion [22], cooperative multi-defender guidance [23], norm-bounded control effort [24], and terminal-angle or finite-time constraints [25, 26]. Zhu *et al.* [25], for instance, develop an event-triggered finite-time cooperative guidance law whose objective decomposes into per-interceptor terminal conditions, namely miss distance and a prescribed impact angle, which are additive across engagements. To move beyond the few-player regime, game-theoretic analysis has recently been cast in the mean-field limit, covering swarm guidance toward macroscopic objectives [11], for many-versus-many pursuit-evasion [27], and for large-scale swarm-versus-swarm attack-defense evolution [12]. These swarm-scale games are nonetheless dynamic and closed-loop, and optimize additive or energy-type cost functionals accumulated along trajectories. Our setting departs from all of the above not in scale but in the structure of the objective, since the collective payoff is the *union*, not the sum, of the per-cloud coverage windows, so the per-engagement decomposition on which these guidance and game formulations rest no longer holds.

A complementary body of work orchestrates UAV swarms through multi-agent task allocation, using consensus-based bundle algorithms [13,28,29], the Contract Net Protocol [30,31], and market or auction mechanisms [14, 32]. These methods scale well precisely because they assume the global objective is decomposable, treating system utility as an aggregate of per-agent contributions. Wang *et al.* [33],

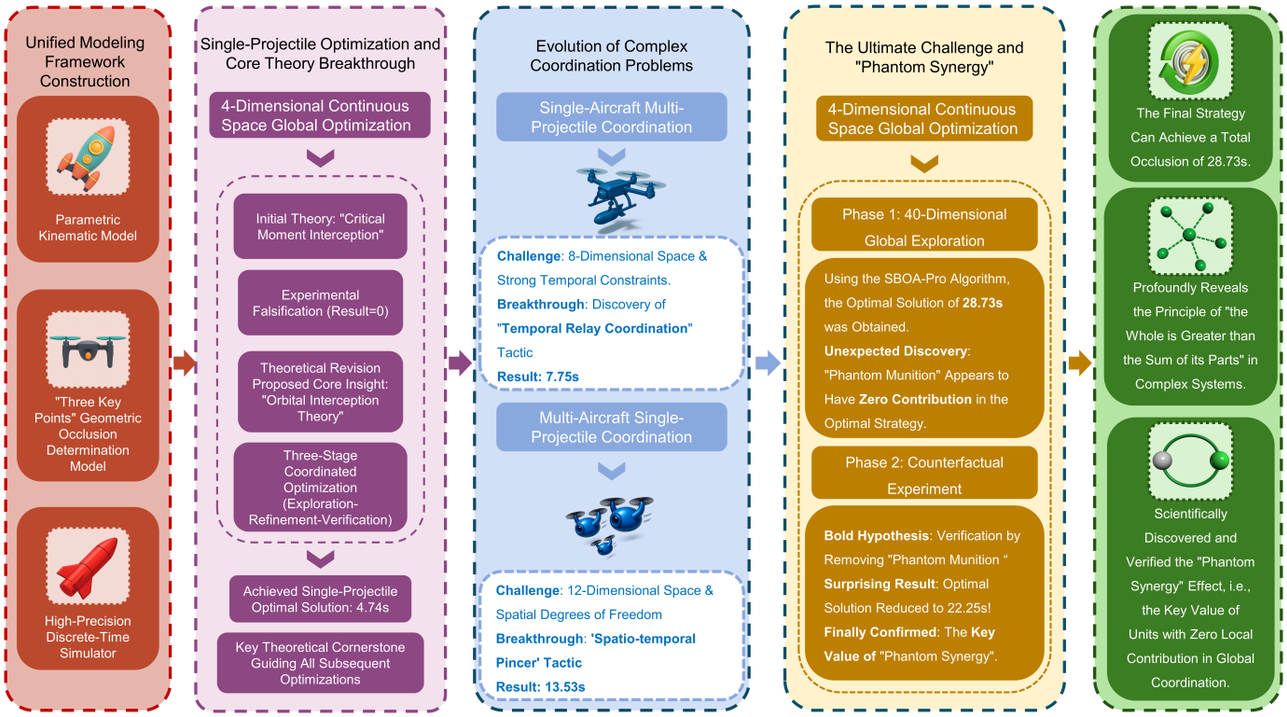


Fig. 1. A progressive technical roadmap for problem-solving. Our research framework follows a three-tiered logic, moving from a foundational layer consisting of a unified kinematics and obscuration model, a strategy evolution layer that progressively scales from single-agent optimization to distributed multi-agent cooperation, and a final integration layer where the full 40-dimensional problem is addressed, where the Phantom Synergy effect is observed.

for example, employ a two-stage greedy auction in which system value is the sum of individual damage advantage values, an additivity that keeps the auction tractable. Heterogeneous unmanned swarms may also cooperate across sea, air, and land, but such surveys remain at roadmap granularity, organizing cooperation around allocating separable subtasks to platforms rather than around a jointly defined, nonseparable payoff [34]. This decomposability fails for our problem, because total obscuration time is the logical union of coverage windows, so no cloud has a value independent of where and when the others are placed. In this regime, agent contributions are complementary rather than additive, and greedy or additive allocation is prone to poor local optima.

2.2. *Coverage maximization and submodularity*

A separate line of work studies coverage and observation-selection problems, in which a subset of sensors, locations, or interventions is chosen to maximize a coverage or information objective. Such objectives are frequently *submodular*, exhibiting diminishing returns, so that adding an element to a smaller set helps at least as much as adding it to a larger one. Submodularity is what renders these otherwise NP-hard subset-selection problems tractable, since a greedy algorithm then attains a $(1 - 1/e)$ approximation of the optimum [35]. This property underlies near-optimal sensor placement in Gaussian processes [36], observation selection for robotic exploration and diagnosis [37], the trade-off between information and communication cost in wireless sensor networks [38], influence maximization in social networks [39], and more recent placement problems for traffic networks [40] and intelligent reflecting surfaces [41]. Even when an objective departs from strict submodularity, greedy selection retains guarantees that degrade gracefully with the submodularity ratio γ, a scalar measuring proximity to submodularity, for which the approximation factor is $1 - e^{-\gamma}$ [42–44].

Our obscuration objective is itself a coverage objective, the union of per-cloud occlusion windows, which would suggest that it inherits this structure together with the greedy guarantee. It does not. Blocking a line of sight requires *all* critical keypoints to be occluded simultaneously, a conjunction that couples clouds so that one cloud can raise the marginal value of another. This complementarity makes the induced set function monotone but nonsubmodular (Proposition 3.4), and drives its submodularity ratio to zero, so even the relaxed guarantees for approximately submodular maximization vanish. Greedy and auction allocations therefore carry no near-optimality guarantee for this problem, and joint optimization becomes necessary.

2.3. *Obscurant deployment and alternative approaches*

The physical basis of smoke-screen defense is well established. A substantial literature characterizes multispectral obscurant compositions, typically red-phosphorus-based formulations, and their attenuation across the visible, infrared, and laser bands [45–48], together with empirical and analytical models of the screening time and shielding area of a single cloud [49, 50]. This work supplies the physical justification for treating a deployed cloud as an effective occluder. It is, however, almost exclusively concerned with the chemistry and radiometry of an *individual* charge, addressing how much a single cloud attenuates, for how long, and over what area. The complementary question, namely how a *swarm* of clouds should be positioned and timed so that their combined coverage blocks every critical line of sight for as long as possible, is a cooperative spatiotemporal planning problem that this material-science literature does not address. Our work takes the single-cloud occlusion model as a physical primitive and studies precisely this collective deployment layer.

Collective behavior can also be pursued through learning instead of explicit planning. The study of emergent behavior often turns to Multi-Agent Reinforcement Learning (MARL) or bio-inspired heuristics [51, 52]. MARL offers potential scalability but faces challenges in safety-critical defense applications, often struggling to guarantee hard geometric constraints and lacking the interpretability required for tactical analysis [53–55]. Our work takes a complementary approach. Instead of training black-box policies or designing local rules in the hope that useful behavior emerges, we compute an optimized solution to the collective task and then analyze it directly. Because the optimized plan is fully specified and open to inspection, the cooperative structure that a learned policy would leave implicit can be recovered and explained from the plan itself.

3. Methodology

3.1. *Problem formulation*

The problem is formulated as a large-scale optimal control problem. Given the initial states of the incoming missiles, the initial positions of the UAV agents, and the physical parameters of the smoke munitions, the task is to determine an open-loop plan, namely the constant flight heading and speed of each agent together with the deployment and fuse timing of each of its munitions, that maximizes the total duration over which the protected asset is effectively obscured from all threats. Let the set of UAV agents be $\mathcal{J} = \{1, \dots, J\}$ and the set of available munitions per agent be $\mathcal{K} = \{1, \dots, K\}$. For convenience, all symbols used throughout this section are summarized in Table 1.

The strategy for the swarm is defined by a decision vector $\vec{X}$, which is the concatenation of all individual agent strategies, $\vec{X} = [\vec{x}_1, \dots, \vec{x}_J]$, residing in a high-dimensional space $\mathbb{R}^D$. In our full-scale scenario, $D = 40$. The strategy vector $\vec{x}_j$ of each agent is composed of its flight parameters and the timing for each of its K munitions:

$$\vec{x}_j = [\theta_j, v_j, t_{\text{drop},j,1}, \Delta t_{\text{fuse},j,1}, \dots, t_{\text{drop},j,K}, \Delta t_{\text{fuse},j,K}],$$

where θ_j is the flight direction, v_j is the speed, $t_{\text{drop},j,k}$ is the deployment time of munition k, and $\Delta t_{\text{fuse},j,k}$ is its fuse delay. Each agent thus contributes two flight parameters and $2K$ munition-timing parameters, i.e. $2 + 2K$ dimensions; with $K = 3$ munitions this is 8 per agent, so the full-scale swarm of $J = 5$ agents gives $D = 40$. During optimization the absolute deployment times $t_{\text{drop},j,k}$ are equivalently re-parameterized as a first deployment time plus successive positive intervals (Sec. 3.4), which turns the ordering constraint of Ω into simple box bounds without changing the feasible set.

The optimization is performed over a feasible decision space Ω, which is formally defined by a set of operational constraints:

$$\Omega = \left\{ \vec{X} \;\middle|\; \begin{array}{l} v_j \in [v_{\min}, v_{\max}], \quad \forall j \in \mathcal{J} \\ t_{\text{drop},j,k+1} - t_{\text{drop},j,k} \geq \Delta t_{\text{interval}}, \\ \qquad \forall j \in \mathcal{J}, k \in \{1, \dots, K-1\} \\ t_{\text{drop},j,k} > 0, \quad \Delta t_{\text{fuse},j,k} > 0, \quad \forall j, k \\ z(\vec{P}_{\text{det},j,k}) \geq 0, \quad \forall j, k \end{array} \right\},$$

where $[v_{\min}, v_{\max}]$ are the lower and upper bounds on the agent flight speed, $\Delta t_{\text{interval}}$ is the minimum time separation enforced between two consecutive munition deployments of the same agent, and $z(\cdot)$ denotes the z-coordinate of a position vector, ensuring above-ground detonation.

The objective function, $T_{\text{total}}(\vec{X})$, is the total time during which at least one missile is effectively obscured. This is formally expressed as the integral of a system-level obscuration indicator function, $I(t; \vec{X})$, over the simulation time horizon T_{sim}:

$$\max_{\vec{X} \in \Omega} T_{\text{total}}(\vec{X}) = \int_0^{T_{\text{sim}}} I(t; \vec{X})\, dt. \tag{1}$$

Here, T_{sim} is a fixed integration bound, chosen large enough to contain the entire engagement. The integrand $I(t; \vec{X})$ can be nonzero only while a missile is still in flight *and* a deployed cloud is simultaneously active. Each cloud contributes over the bounded window $[t_{\text{det},j,k}, t_{\text{det},j,k} + T_{\text{eff}}]$ of Eq. (4), and each missile m reaches its aimpoint at a finite impact time $t_{\text{impact},m} = \|\vec{P}_{\text{aim}} - \vec{P}_{M,0}\|/v_M$, beyond which it no longer threatens the asset. The support of $I(\cdot; \vec{X})$ is therefore contained in the fixed interval $[0, \max_m t_{\text{impact},m} + T_{\text{eff}}]$; any T_{sim} chosen at or beyond this value yields identical T_{total} by construction, since the integrand is identically zero beyond the support. In our full-scale scenario, the longest missile flight

Table 1. Notation used in the problem formulation and system model (Sec. 3).

Symbol	Description
Sets and indices	
$\mathcal{J} = \{1, \dots, J\}$	Set of UAV agents; J is the number of agents.
$\mathcal{K} = \{1, \dots, K\}$	Set of munitions carried per agent; K is the number per agent.
$\mathcal{M}$	Set of incoming missile threats; index m.
$\mathcal{P}_{\text{asset}}$	Set of critical keypoints on the protected asset; index p.
j, k	Indices of a UAV agent and one of its munitions, respectively.
D	Dimension of the full decision space ($D = 40$ in the full-scale scenario).
Decision variables	
$\vec{X}$	System-wide strategy vector, concatenation of all $\vec{x}_j$; $\vec{X} \in \mathbb{R}^D$.
$\vec{x}_j$	Strategy vector of agent j.
$\theta_j,\ v_j$	Flight direction and speed of agent j.
$t_{\text{drop},j,k}$	Deployment (drop) time of munition k of agent j.
$\Delta t_{\text{fuse},j,k}$	Fuse delay of munition (j,k); detonation at $t_{\text{det},j,k} = t_{\text{drop},j,k} + \Delta t_{\text{fuse},j,k}$.
Feasible region	
Ω	Feasible decision space defined by the operational constraints.
$[v_{\min}, v_{\max}]$	Lower/upper bounds on agent flight speed.
$\Delta t_{\text{interval}}$	Minimum time separation between consecutive drops of the same agent.
$z(\cdot)$	z-coordinate (altitude) of a position vector.
Kinematics	
$\vec{P}_M(t),\ \vec{P}_{M,0},\ \vec{P}_{\text{aim}},\ v_M$	Missile position at t, initial position, aimpoint, and constant speed.
$\vec{P}_{FY,j}(t),\ \vec{P}_{FY,j,0},\ \vec{v}_{FY,j}$	UAV agent j position at t, initial position, and constant velocity vector.
$\vec{P}_{\text{det},j,k}$	Detonation position of munition (j,k).
$\vec{P}_{\text{asset},p}$	Position of critical keypoint p on the protected asset.
$\vec{g} = [0, 0, -g]^T$	Gravitational acceleration vector; g its magnitude.
$\vec{P}_{C,j,k}(t),\ \vec{v}_{\text{sink}} = [0, 0, -v_{\text{sink}}]^T$	Smoke-cloud center at t and its constant sinking velocity.
$R_{\text{cloud}},\ T_{\text{eff}}$	Effective radius and effective lifetime of a smoke cloud.
Objective and obscuration logic	
$T_{\text{total}}(\vec{X})$	Total duration of effective obscuration (objective to maximize).
T_{sim}	Simulation time horizon over which T_{total} is integrated.
$I(t; \vec{X})$	System-level obscuration indicator (1 if any missile obscured, else 0).
$I_m(t; \vec{X})$	Obscuration indicator for a single missile m.
$C_{m,j,k,p}(t; \vec{X})$	Binary interception test: 1 if cloud (j,k) blocks the LOS from missile m to keypoint p at t.
$\vec{P}_{\text{closest}},\ \lambda$	Closest point on the LOS segment to the cloud center $\vec{C} \equiv \vec{P}_{C,j,k}(t)$, and its projection parameter.
$\vee,\ \wedge$	Logical OR and AND operators.

time is ≈ 67 s (a 300 m/s threat closing over roughly 20 km) and the cloud lifetime is $T_{\text{eff}} = 20$ s, placing this bound well under 90 s. The reported performance is thus a property of the engagement geometry, not an artifact of the simulation horizon. The indicator function is defined hierarchically. Let $\mathcal{M}$ be the set of missiles and $\mathcal{P}_{\text{asset}}$ be the set of critical points on the protected asset. System-level obscuration at time t occurs if any missile is obscured:

$$I(t; \vec{X}) = \bigvee_{m \in \mathcal{M}} I_m(t; \vec{X}), \tag{2}$$

where $\vee$ denotes the logical OR operator. A single missile m is considered obscured only if the lines-of-sight (LOS) to all critical points are simultaneously intercepted. This requires that for each critical point $p \in \mathcal{P}_{\text{asset}}$, at least one munition (j,k) is blocking the corresponding LOS:

$$I_m(t; \vec{X}) = \bigwedge_{p \in \mathcal{P}_{\text{asset}}} \left(\bigvee_{j \in \mathcal{J}} \bigvee_{k \in \mathcal{K}} C_{m,j,k,p}(t; \vec{X}) \right). \tag{3}$$

The critical-point set $\mathcal{P}_{\text{asset}}$ discretizes the silhouette of the protected asset for the line-of-sight test. We model the asset as an axis-symmetric vertical body and take $\mathcal{P}_{\text{asset}} = \{\text{top}, \text{center}, \text{bottom}\}$, the two extremal points and the mid-point spanning its vertical extent (Fig. 2). Requiring all three keypoints to be screened *simultaneously* is a more demanding criterion than testing a single centroidal point, because adding sample points along the silhouette can only shorten the measured obscuration and never lengthen it, so the three-point model reports a conservative lower bound on what a single-point test would credit, while remaining cheap

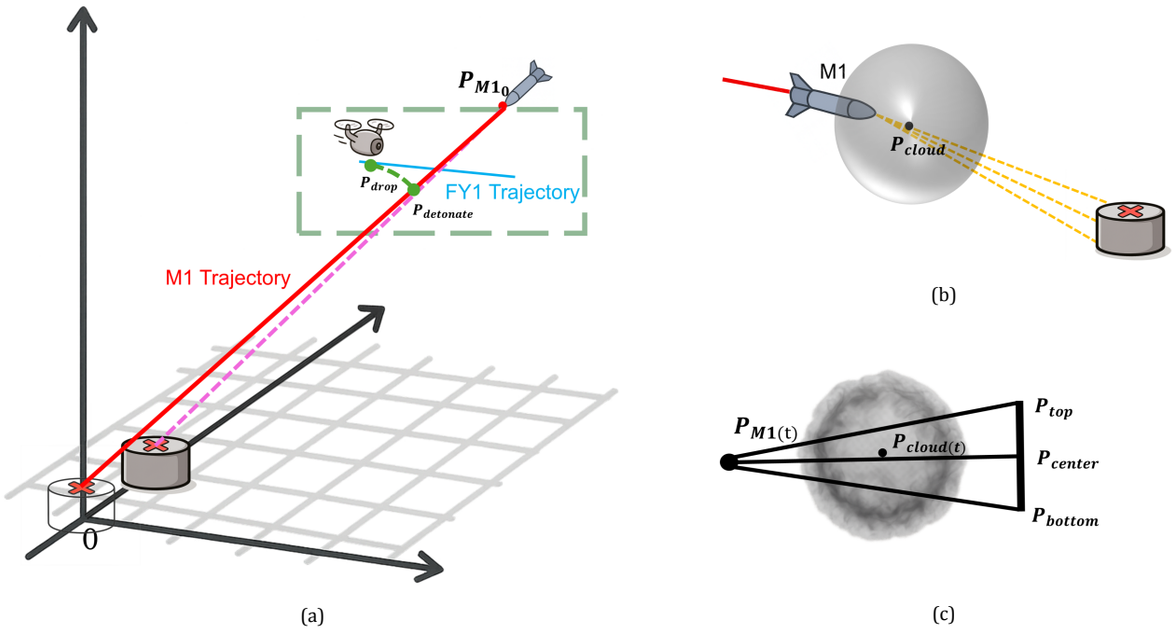


Fig. 2. Illustration of the three-key-points geometric obscuration criterion. (a) A panoramic view of the engagement, showing the trajectories of the UAV, missile, and smoke munition. The highlighted segment on the path of the missile indicates the time of effective obscuration. (b) A close-up view of the geometry at the moment of obscuration. (c) The schematic demonstrates that effective obscuration is achieved only when the lines of sight from the missile to all three critical points on the protected asset (top, center, bottom) are simultaneously intercepted by the spherical smoke cloud.

enough to evaluate over the entire population and horizon. It captures the vertical envelope a seeker would exploit, and the formulation extends verbatim to a finer keypoint set at a proportional increase in evaluation cost. Here, $\wedge$ is the logical AND operator, and $C_{m,j,k,p}(t;\vec{X})$ is a binary function evaluating the fundamental interception condition. This condition is met if the smoke cloud from munition (j,k) intercepts the LOS from missile m to point p at time t. Formally, this geometric test is defined as

$$C_{m,j,k,p}(t;\vec{X}) = \begin{cases} 1 & \text{if } \|\vec{C} - \vec{P}_{\text{closest}}\| \le R_{\text{cloud}} \\ & \text{and } t \in [t_{\text{det},j,k},\, t_{\text{det},j,k} + T_{\text{eff}}], \\ 0 & \text{otherwise,} \end{cases} \tag{4}$$

where $\vec{C} = \vec{P}_{C,j,k}(t)$ is the smoke cloud's center, R_{cloud} is its radius, and T_{eff} is its effective lifetime. The point $\vec{P}_{\text{closest}}$ on the LOS segment is found by projecting $\vec{C}$ onto the line defined by the missile position $\vec{A} = \vec{P}_M(t)$ and asset keypoint $\vec{B} = \vec{P}_{\text{asset},p}$:

$$\vec{P}_{\text{closest}} = \vec{A} + \max(0, \min(1, \lambda))(\vec{B} - \vec{A}),$$
$$\text{with } \lambda = \frac{(\vec{C} - \vec{A}) \cdot (\vec{B} - \vec{A})}{\|\vec{B} - \vec{A}\|^2}. \tag{5}$$

The explicit kinematic equations for calculating the time-dependent positions $\vec{P}_M(t)$, $\vec{P}_{C,j,k}(t)$, and $\vec{P}_{\text{det},j,k}$ as functions of the decision vector $\vec{X}$ are detailed in the following section.

3.2. *System model*

To evaluate the objective $T_{\text{total}}(\vec{X})$ for a given decision vector $\vec{X}$, we use a deterministic kinematic model that maps a plan to the trajectories of every entity in the engagement. The model resolves the engagement at the level of the planning problem, that is, the positions, timings, and geometry that determine obscuration, and treats the low-level dynamics of each platform as an underlying layer summarized by a small set of assumptions, stated next. It serves as the black-box objective that the optimizer maximizes, supplying the positions required in Sec. 3.1.

The kinematic model rests on three assumptions that fix the scope of the study.

Missile threat. Each incoming missile is modeled as following a constant-speed rectilinear trajectory toward a designated aimpoint, given by Eq. (6). The guidance law of the missile is not modeled. The missile approach is abstracted as a known deterministic threat trajectory against which the defense is planned, and it is this trajectory that the line-of-sight geometry of Sec. 3.1 acts upon. The threat speed v_M, aimpoint $\vec{P}_{\text{aim}}$, and initial state $\vec{P}_{M,0}$ are inputs to the problem, not decision variables.

UAV control. Each agent is commanded at the planning level by a constant flight heading θ_j and speed v_j, together with its munition deployment and fuse schedule, which are precisely the decision variables of $\vec{x}_j$. The resulting horizontal

constant-velocity motion is given by Eq. (7). A lower-level flight-control system is assumed to track these constant-velocity commands, and designing that autopilot is outside the present scope.

Planning horizon and feedback. The formulation is an open-loop cooperative planning problem, in which the entire heading, speed, and deployment schedule is committed at the outset and executed without in-flight revision. Feedback-loop stability is therefore a property of the underlying flight controller or of a future receding-horizon, closed-loop extension (Sec. 5), and lies one layer below the planning problem solved here. Throughout this paper, cooperative control denotes this high-level cooperative planning and scheduling of the swarm, not a new closed-loop feedback control law.

The system dynamics are described within a unified three-dimensional Cartesian coordinate system. The trajectory of each entity is governed by a deterministic kinematic model.

The position of a missile m at time t, denoted $\vec{P}_M(t)$, is modeled as linear motion from its initial position $\vec{P}_{M,0}$ toward an aimpoint $\vec{P}_{\text{aim}}$ at a constant speed v_M:

$$\vec{P}_M(t) = \vec{P}_{M,0} + v_M \frac{\vec{P}_{\text{aim}} - \vec{P}_{M,0}}{\|\vec{P}_{\text{aim}} - \vec{P}_{M,0}\|} t. \tag{6}$$

The position of a UAV agent j, $\vec{P}_{FY,j}(t)$, is determined by its initial position $\vec{P}_{FY,j,0}$ and its constant velocity vector $\vec{v}_{FY,j}$. This velocity is directly constructed from the decision variables v_j and θ_j within the strategy vector $\vec{x}_j$:

$$\vec{P}_{FY,j}(t) = \vec{P}_{FY,j,0} + \vec{v}_{FY,j} t,$$
$$\text{where } \vec{v}_{FY,j} = [v_j \cos\theta_j, v_j \sin\theta_j, 0]^T. \tag{7}$$

The motion of a munition (j,k) is piecewise. Deployed at the decision time $t_{\text{drop},j,k}$, it follows a projectile trajectory under gravity $\vec{g} = [0,0,-g]^T$ until detonation at $t_{\text{det},j,k} = t_{\text{drop},j,k} + \Delta t_{\text{fuse},j,k}$, where $\Delta t_{\text{fuse},j,k}$ is also a decision variable. Its detonation position $\vec{P}_{\text{det},j,k}$ is given by

$$\vec{P}_{\text{det},j,k} = \vec{P}_{FY,j}(t_{\text{drop},j,k}) + \vec{v}_{FY,j}\Delta t_{\text{fuse},j,k} + \frac{1}{2}\vec{g}(\Delta t_{\text{fuse},j,k})^2. \tag{8}$$

For $t \geq t_{\text{det},j,k}$, the center of the resulting spherical smoke cloud, $\vec{P}_{C,j,k}(t)$, descends from the detonation point at a constant sinking velocity $\vec{v}_{\text{sink}} = [0,0,-v_{\text{sink}}]^T$:

$$\vec{P}_{C,j,k}(t) = \vec{P}_{\text{det},j,k} + \vec{v}_{\text{sink}}(t - t_{\text{det},j,k}). \tag{9}$$

3.3. *Analytical properties of the obscuration problem*

The obscuration objective admits four analytical properties that justify the use of derivative-free metaheuristics, account for the qualitative behavior observed in the results, and characterize the structure of the cooperative planning problem.

Proposition 3.1. *The feasible set Ω is nonempty and compact, and the objective $T_{\text{total}}(\vec{X})$ is bounded on Ω.*

Proof. Each decision variable ranges over a closed bounded interval. The flight heading satisfies $\theta_j \in [0, 2\pi]$ and the speed $v_j \in [v_{\min}, v_{\max}]$. Re-parameterizing the absolute deployment times $(t_{\text{drop},j,1}, \ldots, t_{\text{drop},j,K})$ as a first drop time $t_{\text{drop},j,1} \in [0, \bar{t}]$ followed by $K-1$ successive intervals $\delta_{j,k} \in [\delta_{\min}, \delta_{\max}]$ encodes the temporal ordering constraint $t_{\text{drop},j,k} < t_{\text{drop},j,k+1}$ and the minimum separation directly as box bounds (Sec. 3.4), and each fuse delay obeys $\Delta t_{\text{fuse},j,k} \in [0, \overline{\Delta t}]$. The feasible set Ω is therefore contained in a closed bounded box $B \subset \mathbb{R}^D$. Every remaining constraint, for instance the altitude condition on each detonation point, is a nonstrict inequality on a continuous function of $\vec{X}$, so its solution set is closed and Ω is the intersection of these closed sets with B, hence closed. A closed subset of the bounded box B is both closed and bounded, and is therefore compact by the Heine–Borel theorem. Nonemptiness follows by construction, since assigning every agent an admissible heading and speed, spacing its drops by $\delta_{\min}$ from $t_{\text{drop},j,1} = 0$, and taking a fuse delay small enough that the projectile falls only a short distance before detonation keeps every detonation point above ground, so all constraints are met and $\Omega \neq \emptyset$. The integrand satisfies $I(t;\vec{X}) \in \{0,1\}$ and the horizon T_{sim} is finite, so $0 \leq T_{\text{total}}(\vec{X}) = \int_0^{T_{\text{sim}}} I(t;\vec{X})\, dt \leq T_{\text{sim}}$ for every $\vec{X} \in \Omega$, and the objective is bounded. □

Proposition 3.2. *Let a spherical smoke cloud of radius R_{cloud} translate with velocity $\vec{v}_C$ in the presence of a missile with velocity $\vec{v}_M$, and let $\hat{u}$ be the unit line-of-sight (LOS) direction at the window center, taken constant over the interception window. Let the relative velocity perpendicular to the LOS be*

$$\vec{b} = (\vec{v}_C - \vec{v}_M) - [(\vec{v}_C - \vec{v}_M) \cdot \hat{u}]\, \hat{u}$$

and let $d_{\min}$ be the minimum perpendicular distance from the cloud trajectory to the LOS. If $d_{\min} < R_{\text{cloud}}$, the duration Δ during which the cloud intercepts the LOS is

$$\Delta = \frac{2}{\|\vec{b}\|}\sqrt{R_{\text{cloud}}^2 - d_{\min}^2}. \tag{10}$$

The constant-$\hat{u}$ approximation holds when $\Delta \ll t_{\text{impact}}$, so that the LOS rotates negligibly during interception, and the closest point is taken interior to the LOS segment throughout the window.

Proof. In the reference frame of the missile, the cloud center moves along the straight line $\vec{C}(\tau) = \vec{C}(0) + (\vec{v}_C - \vec{v}_M)\tau$, where $\tau = 0$ denotes the instant of closest approach to the

Table 2. Validation of the obscuration window formula, Eq. (10), for a single cloud against missile M1 (Q_2 geometry, $R_{\text{cloud}} = 10$ m). The higher relative error at $v_{\text{sink}} = 0$ arises from temporal discretization in the simulation when the cloud is approximately stationary relative to the LOS; errors fall below 1% for all nonzero sink rates.

v_{sink} (m/s)	Sim. Δ (s)	Anal. Δ (s)	Error	Note
0.0	3.03	3.19	5.3%	Static
1.0	5.10	5.09	0.2%	
1.5	5.36	5.33	0.6%	Peak
2.0	5.31	5.27	0.8%	
3.0	4.71	4.67	0.9%	Physical
6.0	2.71	2.72	0.4%	

LOS. The relative velocity $\vec{v}_C - \vec{v}_M$ decomposes into a component along $\hat{u}$ and the perpendicular component $\vec{b}$. Motion along $\hat{u}$ displaces the cloud center only in the LOS direction and leaves its perpendicular distance to the LOS unchanged, so the perpendicular displacement $\vec{r}(\tau)$ from the LOS to $\vec{C}(\tau)$ evolves as $\vec{r}(\tau) = \vec{r}(0) + \vec{b}\,\tau$. At closest approach $\vec{r}(0)$ is orthogonal to $\vec{b}$ with $\|\vec{r}(0)\| = d_{\text{min}}$, so by the Pythagorean theorem the perpendicular deviation is

$$\|\vec{r}(\tau)\| = \sqrt{d_{\text{min}}^2 + \|\vec{b}\|^2\tau^2}.$$

The cloud intercepts the LOS while this deviation does not exceed the cloud radius, that is $\sqrt{d_{\text{min}}^2 + \|\vec{b}\|^2\tau^2} \le R_{\text{cloud}}$. This inequality has solutions only when $d_{\text{min}} < R_{\text{cloud}}$, in which case it is equivalent to $|\tau| \le \sqrt{R_{\text{cloud}}^2 - d_{\text{min}}^2}/\|\vec{b}\|$. The interception interval is therefore symmetric about $\tau = 0$ with half-width $|\tau|_{\text{max}} = \sqrt{R_{\text{cloud}}^2 - d_{\text{min}}^2}/\|\vec{b}\|$, and its total duration is $\Delta = 2|\tau|_{\text{max}}$, which is Eq. (10). □

Equation (10) shows that the obscuration window is a geometric chord determined solely by d_{min} and $\|\vec{b}\|$, independent of the cloud lifetime T_{eff}. The cloud traverses the line of sight in only a few seconds regardless of how long it remains chemically active, which explains why optimized windows of ≈ 4.7 s are far shorter than the 20 s smoke lifetime. Table 2 validates Eq. (10) against the kinematic simulation across a range of sink rates.

The sink rate v_{sink} sets up two competing effects on the obscuration window. A sinking cloud tracks the descending LOS more closely, which reduces the miss distance d_{min} and lengthens the geometric chord $\sqrt{R_{\text{cloud}}^2 - d_{\text{min}}^2}$ through the cloud. The same vertical motion raises the perpendicular sweep rate $\|\vec{b}\|$ across the LOS, which shortens the window. At low sink rates, the miss distance falls steeply while the sweep rate grows only slowly, so the chord gain dominates and the window widens. At high sink rates the sweep rate grows without bound and drives the window to zero, so an interior optimum in v_{sink} is expected.

Proposition 3.3. *Suppose the cloud velocity depends affinely on the sink rate, $\vec{v}_C(v_{\text{sink}}) = \vec{v}_C^{\,0} - v_{\text{sink}}\hat{z}$, that the relative trajectory is never parallel to the LOS, so $\vec{b}(v_{\text{sink}}) \neq 0$ for all $v_{\text{sink}} \ge 0$, and that $d_{\text{min}}(0) < R_{\text{cloud}}$. Then the obscuration window $\Delta(v_{\text{sink}})$ of Eq. (10) is continuous on $[0,\infty)$ with $\Delta(v_{\text{sink}}) \to 0$ as $v_{\text{sink}} \to \infty$. If in addition $\Delta(v_1) > \Delta(0)$ for some $v_1 > 0$, then Δ attains its maximum at an interior point $v_{\text{sink}}^\star \in (0,\infty)$, so the optimal sink rate is finite and strictly positive.*

Proof. Because $\vec{v}_C$ is affine in v_{sink}, so is the relative velocity $\vec{w} = \vec{v}_C - \vec{v}_M$ and its perpendicular component $\vec{b}$. Hence $\|\vec{b}(v_{\text{sink}})\|$ is the norm of an affine vector and is continuous and positive, using $\vec{b} \neq 0$. In the missile frame, the cloud center travels along the line $\{\vec{C}_0 + \vec{w}\tau\}$ while the LOS is the line $\{s\hat{u}\}$, so d_{min} is the distance between the two lines,

$$d_{\text{min}}(v_{\text{sink}}) = \frac{|\vec{C}_0 \cdot (\vec{w} \times \hat{u})|}{\|\vec{w} \times \hat{u}\|}.$$

Both $\vec{C}_0 \cdot (\vec{w} \times \hat{u})$ and $\vec{w} \times \hat{u}$ are affine in v_{sink}, and the denominator is nonzero because $\vec{b} \neq 0$ excludes parallelism, so d_{min} is continuous. The physical window is $\Delta(v_{\text{sink}}) = 2\,g(d_{\text{min}}(v_{\text{sink}}))/\|\vec{b}(v_{\text{sink}})\|$ when $d_{\text{min}} < R_{\text{cloud}}$ and $\Delta = 0$ when $d_{\text{min}} \ge R_{\text{cloud}}$, since the cloud then never blocks the LOS. With $g(d) = \sqrt{R_{\text{cloud}}^2 - d^2}$ continuous on $[0, R_{\text{cloud}}]$ and $g(R_{\text{cloud}}) = 0$, the two pieces agree at $d_{\text{min}} = R_{\text{cloud}}$, so Δ is continuous on $[0,\infty)$.

As $v_{\text{sink}} \to \infty$, $\vec{w} = \vec{v}_C^{\,0} - \vec{v}_M - v_{\text{sink}}\hat{z}$ grows without bound in the $\hat{z}$ direction, which is not parallel to $\hat{u}$, so $\|\vec{b}(v_{\text{sink}})\| \to \infty$. Since $0 \le g(d_{\text{min}}) \le R_{\text{cloud}}$ stays bounded, $\Delta \to 0$.

Suppose $\Delta(v_1) > \Delta(0)$ for some $v_1 > 0$. Because $\Delta \to 0$, there exists $M > v_1$ with $\Delta(v_{\text{sink}}) < \Delta(v_1)$ for all $v_{\text{sink}} \ge M$. On the compact interval $[0, M]$ the continuous function Δ attains a maximum at some $v_{\text{sink}}^\star$. This maximizer is not $v_{\text{sink}} = 0$, since $\Delta(v_1) > \Delta(0)$, and not $v_{\text{sink}} = M$, since $\Delta(M) < \Delta(v_1)$, so $v_{\text{sink}}^\star \in (0, M)$. As $\Delta(v_{\text{sink}}) < \Delta(v_1) \le \Delta(v_{\text{sink}}^\star)$ for all $v_{\text{sink}} > M$, this is also the global maximizer on $[0,\infty)$. □

The substantive part of the claim is the rise hypothesis $\Delta(v_1) > \Delta(0)$, since continuity and the decay to zero are structural. This hypothesis is confirmed numerically for the single-agent optimum against missile M1. As v_{sink} increases from 0 to 1.5 m/s, the measured miss distance falls from $d_{\text{min}} \approx 6.98$ m to 3.05 m, lengthening the chord by about a third, while the sweep rate $\|\vec{b}\|$ rises by only about a tenth. The analytical window of Eq. (10) then peaks at $v_{\text{sink}} \approx 1.5$ m/s with $\Delta \approx 5.36$ s, a 77% improvement over the

static window and in agreement with the direct simulation in Table 2. The physical sink rate $v_{\text{sink}} = 3.0$ m/s is a fixed munition property and is not a decision variable. The optimizer instead selects the detonation position and timing to minimize $d_{\min}$, approaching the geometric optimum within the given kinematics. This is the mechanism underlying the orbital interception tactic identified in the single-agent results (Sec. 4).

The objective $T_{\text{total}}(\vec{X})$ has three structural properties that constrain the choice of solver. Because the feasible set is compact (Proposition 3.1), a maximizer is guaranteed to exist, so the search is well-posed and the difficulty lies entirely in the shape of the objective. It is *nonconcave* (maximization over a nonconvex feasible set). Its gradient is *approximately zero almost everywhere*, since the integrand is a binary indicator whose switching locus has measure zero in $\mathbb{R}^D$, providing no useful first-order information. With $D = 40$, grid or exhaustive search is also ruled out. Together these properties eliminate gradient-based, convex-relaxation, and enumeration-based methods, leaving derivative-free global metaheuristics as the pragmatic choice. A single-cloud analytical baseline from Eq. (10) validates the resulting pipeline. Three missiles at the single-cloud peak give $3 \times 5.36 \approx 16.1$ s as a representative noncooperative baseline, whereas the full 40D cooperative solution reaches 28.73 s. The cooperative result exceeding this analytical reference is consistent with physical plausibility, and the magnitude of the cooperative gain is quantified in the Discussion.

Proposition 3.4. *Let V denote the set of all deployable smoke clouds, and define $f : 2^V \to \mathbb{R}_{\geq 0}$ as the total obscuration time achieved by activating a subset $S \subseteq V$. Then f is monotone ($S \subseteq T \Rightarrow f(S) \leq f(T)$) but not submodular, since there exist $S \subseteq T \subseteq V$ and $e \in V \setminus T$ such that*

$$f(S \cup \{e\}) - f(S) < f(T \cup \{e\}) - f(T). \tag{11}$$

Consequently, greedy selection carries no standard $(1 - 1/e)$ approximation guarantee [35].

Proof. Monotonicity holds first. Fix any keypoint p and time t. Adding a cloud e to an active set S can only add to the set of clouds that cover the LOS to p at t, so the disjunction $\bigvee_{j,k} C_{m,j,k,p}$ in Eq. (4) is preserved or newly satisfied, never removed. Consequently the conjunction over all three keypoints, and hence the set of instants at which the full LOS is blocked, can only enlarge. Since Lebesgue measure is monotone under set inclusion, integrating the indicator over $[0, T_{\text{sim}}]$ gives $S \subseteq T \Rightarrow f(S) \leq f(T)$, so f is monotone.

Submodularity fails on the following instance, which is realizable because each cloud covers a keypoint through an independent geometric condition and the keypoints are distinct. The asset defines three keypoints. Let cloud A block the LOS to keypoints $\{1, 2\}$ but leave keypoint 3 exposed over a common time window of positive measure, and let cloud B block only keypoint 3 over that window. Because the obscuration criterion is a conjunction requiring all three keypoints to be blocked simultaneously, neither cloud alone blocks the full LOS, so $f(\{A\}) = f(\{B\}) = 0$. Together, A and B cover all three keypoints over the window, so $f(\{A, B\}) = \tau > 0$ equals the measure of the window. With $S = \emptyset$, $T = \{A\}$, and $e = B$,

$$f(S \cup \{e\}) - f(S) = f(\{B\}) - f(\emptyset) = 0,$$

while

$$f(T \cup \{e\}) - f(T) = f(\{A, B\}) - f(\{A\}) = \tau > 0,$$

so the strict inequality in (11) holds and submodularity fails. The marginal contribution of B *rises* with the context $\{A\}$, the defining signature of complementarity rather than diminishing returns.

The same instance drives the submodularity ratio to zero. That ratio is the largest $\gamma \geq 0$ such that $\sum_{e \in U \setminus L} [f(L \cup \{e\}) - f(L)] \geq \gamma [f(L \cup U) - f(L)]$ for all L and U [42]. Taking $L = \emptyset$ and $U = \{A, B\}$, the left side is $f(\{A\}) + f(\{B\}) = 0$ while the right side is $\gamma f(\{A, B\}) = \gamma\tau$ with $\tau > 0$, which forces $\gamma = 0$. The greedy guarantee for approximately submodular maximization, of the form $1 - e^{-\gamma}$ [42, 44], therefore collapses to zero, and the ratio-based guarantees for this problem class also fail. □

This complementarity is the structural precursor of the Phantom Synergy effect analyzed empirically in Sec. 5.

This work addresses the open-loop cooperative planning layer. Lyapunov stability of the swarm is a property of the underlying lower-level flight controller, assumed present but outside the scope of this work, or of a future receding-horizon extension (Sec. 5). Within the planning layer, all trajectories remain bounded: feasible velocities lie in $[v_{\min}, v_{\max}]$ and the simulation horizon is finite. The continuity of Δ in $d_{\min}$ in Eq. (10) further implies that bounded detonation-placement errors degrade the obscuration window gracefully, providing a natural robustness characterization at the planning level.

3.4. *Optimization framework and progressive protocol*

Solving the high-dimensional (40D), nonlinear, and nonconcave optimization problem calls for a derivative-free global optimizer. We adopt the Secretary Bird Optimization Algorithm (SBOA) [56], a recent population-based metaheuristic that requires no gradient information, an advantage given the simulation-based objective function used here. Its bio-inspired mechanics, which model the distinct exploratory and exploitative phases of a hunt, navigate the high-dimensional decision landscape. As a metaheuristic, SBOA does not guarantee global optimality, so the solu-

tions presented in this paper are interpreted as near-optimal strategies.

The standard SBOA was adapted to the scale of the problem and to reduce premature convergence. We developed an enhanced version, which we term SBOA-Pro, incorporating two problem-specific enhancements. The first is an elite seeding strategy, which leverages insights from the lower-dimensional stages of the analysis by injecting their solutions into the initial population of the full-scale optimization, accelerating convergence toward promising regions of the search space. The second is a flood diversity injection mechanism that counteracts optimization stagnation. If the global best solution fails to improve for a predefined number of iterations, this mechanism resets a substantial portion of the population, forcing the algorithm to escape local optima and explore new, diverse areas of the decision space. A formal description of the algorithm, including its pseudocode and hyperparameter configuration, is provided in Appendix A to ensure full reproducibility. To handle the temporal interval constraints during optimization, we re-parameterized the decision variables from absolute deployment times to a combination of an initial deployment time and subsequent relative intervals, transforming the constraint into simple variable bounds.

This optimizer is applied within a progressive multi-stage protocol, where each stage uses a purpose-built methodology to isolate a distinct source of performance gain. The first stage established a single-agent baseline. To locate a good solution in this tractable 4D space, we designed a Prospecting–Refining–Verifying three-stage hybrid optimization framework that combined large-scale grid search, local refinement, and global verification. The second stage was designed to isolate purely temporal synergy by analyzing a single agent with multiple munitions under a strict temporal constraint, revealing the emergence of temporal relay tactics. As the problem scaled to multi-agent coordination in the third stage (12D), we introduced tactical dimensionality reduction based on physical insight. By identifying an agent with a significant geometric advantage and fixing its optimal strategy as a tactical anchor, the problem is reduced to a more tractable 8D search. The fourth stage addressed the complete 40-dimensional problem. Leveraging insights from the preceding stages within the SBOA-Pro framework, this final-stage analysis led to the discovery of nontrivial emergent behaviors, including the Phantom Synergy effect, the empirical signature of the nonsubmodularity established in Sec. 3.

4. Results

4.1. *Staged performance from single agent to full swarm*

Our analysis began by establishing single-agent performance. A naive, predefined strategy, where the UAV flies directly toward a decoy target and deploys its munition at a fixed time, resulted in a total obscuration time of 1.42 s. Applying our optimization framework to a single agent (Stage 1), the best-found strategy achieved 4.74 s, a 233.8% improvement. This gain, illustrated in Fig. 3, demonstrates the value of optimizing even the flight path and timing of a single agent to create a prolonged orbital interception, in which the descending smoke cloud maintains alignment with the line of sight of the missile. This result serves as the Optimized Single-Agent Baseline (OSAB) for all subsequent cooperative analyses.

We then evaluated the performance gains from cooperation. In the constrained temporal cooperation scenario (Stage 2), where a single UAV deploys three munitions with a minimum 1-s interval, the best-found strategy achieved 7.75 s. As shown in Fig. 4, this strategy implements a temporal relay tactic, where the detonation of each successive munition is timed to coincide with the end of the effective period of the previous munition, creating a continuous obscuration corridor. In the distributed spatio-temporal cooperation scenario (Stage 3), involving three independent UAVs, the performance increased to 13.53 s. The best-found solution in this stage, depicted in Fig. 5, leverages a spatio-temporal pincer tactic. The UAVs approach from different spatial locations and time their deployments to cover different segments of the trajectory of the missile, representing a 185.4% improvement over the OSAB and supporting the value of distributed multi-agent cooperation.

In the final full-scale analysis (Stage 4), involving five UAVs and up to fifteen munitions against three missile threats, the planner identified a strategy that yielded a total obscuration time of 28.73 s. Examination of this best-found solution revealed a nontrivial resource allocation. Of the 15 available munitions, the planner deployed eight that contributed directly to the total duration, while the remaining seven deployed munitions had individual effective obscuration times of zero. This finding suggests the presence of a synergistic mechanism that is not apparent from simply summing individual agent contributions, motivating further investigation into the nature of this emergent behavior. A trajectory and deployment overview for this strategy is provided in Fig. 6.

4.2. *Computational complexity and scalability*

A single evaluation of the obscuration objective requires computing the union of smoke-cloud intersection intervals over all munitions and missiles. The dominant cost is the time-stepping loop: for each of the N_U agents, each of their K munitions, and each of the N_M threat trajectories, every time step checks whether all N_p keypoint line-of-sight segments are blocked by the smoke cloud sphere. The per-evaluation

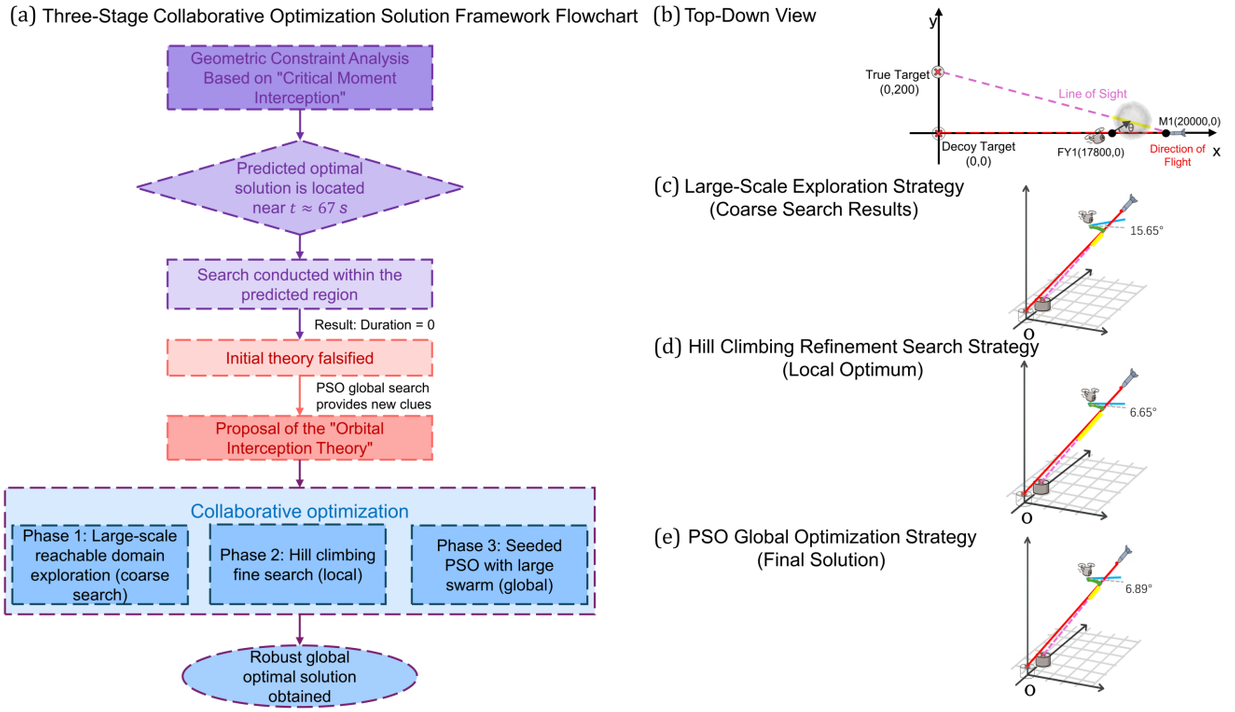


Fig. 3. Visual analysis of the single-agent optimization process. The framework progresses from a large-scale prospecting search that identifies a high-potential solution (4.16 s), to a local refinement stage using Hill Climbing (4.72 s), and finally to a global verification stage with a seeded metaheuristic algorithm that confirms the verified best-found solution (4.74 s). The 2D top-down view shows the subtle yet critical adjustments to the UAV's flight trajectory at each stage.

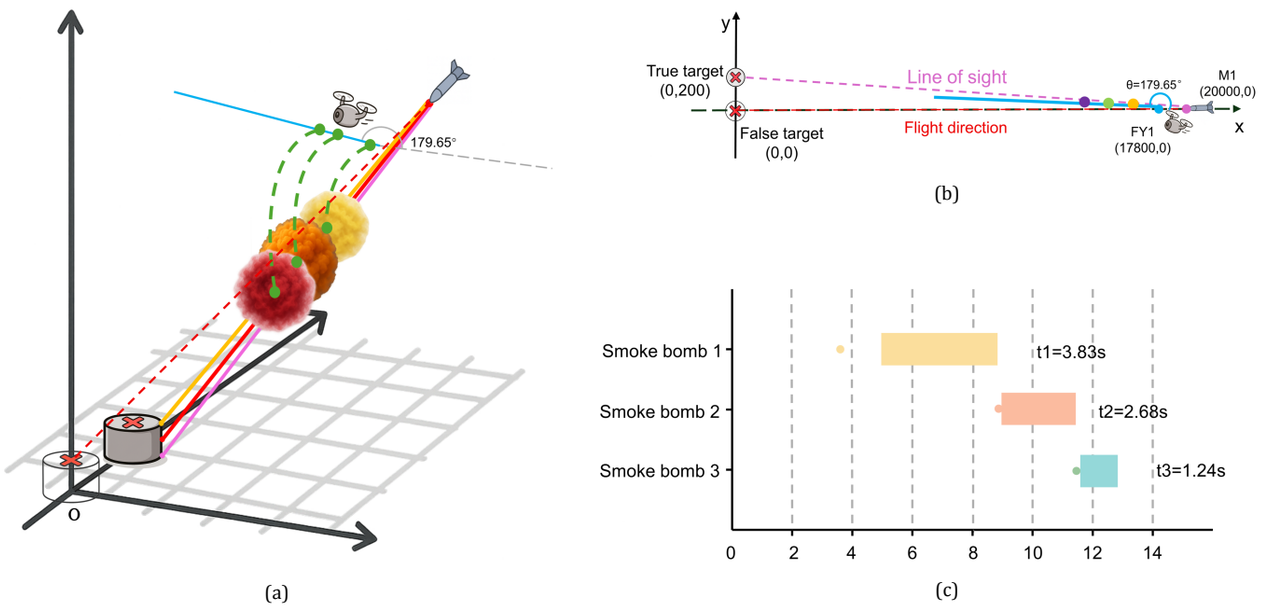


Fig. 4. Best-found strategy for constrained temporal cooperation (Stage 2). A single UAV deploys three munitions along an optimal trajectory. The Gantt chart reveals the core tactic: a seamless *temporal relay* where the three munitions create a continuous obscuration window lasting 7.75 s.

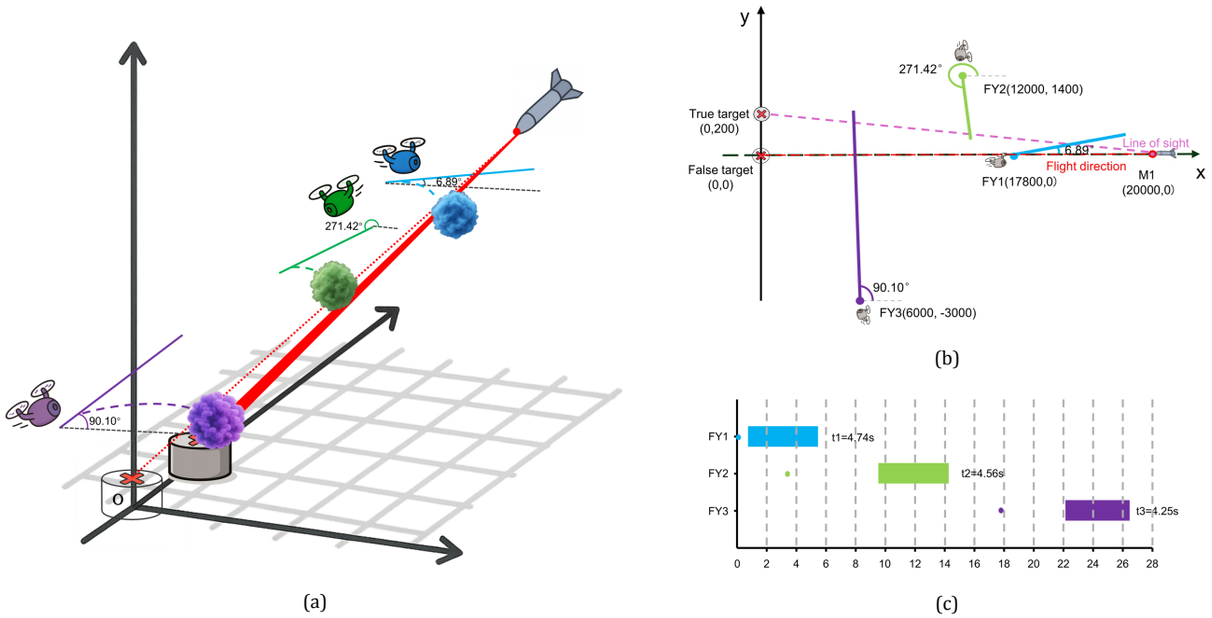


Fig. 5. Best-found strategy for distributed spatio-temporal cooperation (Stage 3). Three UAVs execute a coordinated *pincer* maneuver. The Gantt chart shows that their individual obscuration intervals do not overlap, indicating a highly efficient division of labor over time that results in a total duration of 13.53 s.

complexity is therefore

$$T_{\text{eval}} = \mathcal{O}(N_U \cdot K \cdot N_M \cdot N_t \cdot N_p), \tag{12}$$

where $N_t = T_{\text{smoke}}/\Delta t = 2000$ (at $\Delta t = 0.01\,\text{s}$) and $N_p = |\mathcal{P}_{\text{asset}}| = 3$ are fixed by the physical model. The decision-vector dimension scales as $D = N_U(2 + 2K)$, growing linearly with both N_U and K; for the current configuration this gives $D = 5 \times 8 = 40$. The variable N_M does not appear in the dimension formula. Adding missiles increases the per-evaluation cost (Eq. (12)) but does not expand the decision space.

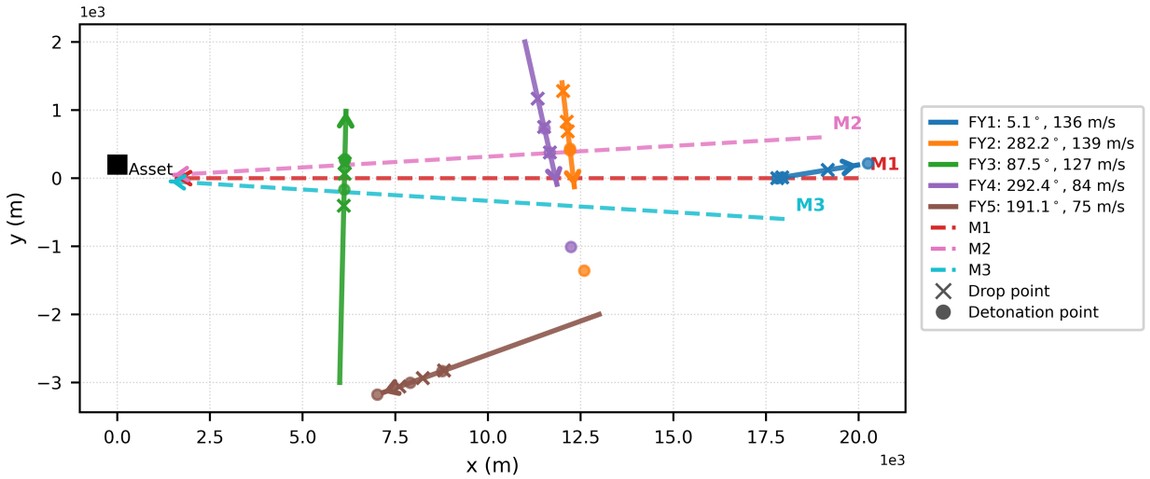
To quantify these scaling trends empirically, we timed the objective evaluation over a 5×3 grid of (N_U, K) configurations at $N_M = 3$ (Table 3), using a reference Python implementation of the objective; the deployed C++ optimizer (compiled with $-$O3) preserves the same asymptotic scaling at substantially lower absolute times. Evaluation time scales near-linearly with both N_U and K (observed ratios $\approx 0.91N_U$ and $\approx 0.96K$, slightly sub-linear due to early-exit keypoint checks), and exactly linearly with N_M (ratio ≈ 2.99 for $N_M = 3$ versus $N_M = 1$).

Two key optimizations reduce constant-factor cost in the C++ implementation. First, all missile trajectories are precomputed once at program start and stored as hash-map lookups, eliminating repeated kinematic arithmetic inside the inner loop. Second, the entire population of $N_{\text{pop}} = 10{,}000$ candidate solutions is evaluated in parallel via OpenMP (capped at 60 threads in our experiments), reducing wall-clock time per generation by up to N_{cores}. In practice, the full 40D optimization ($N_{\text{pop}} = 10{,}000$, $T_{\max} = 50{,}000$ generations, 60 cores) required approximately one day of wall-clock time. This cost is dominated by the scale parameters chosen for thorough exploration of the high-dimensional space, and substantially smaller populations suffice for the lower-dimensional sub-problems in Stages 1–3. Because both the per-evaluation cost and the decision dimension grow only linearly in the agent and munition counts (Eq. 12), the formulation remains tractable at the scale considered here. Substantially larger swarms would shift the bottleneck to population evaluation, which additional engineering such as GPU-accelerated batching or spatial decomposition could address.

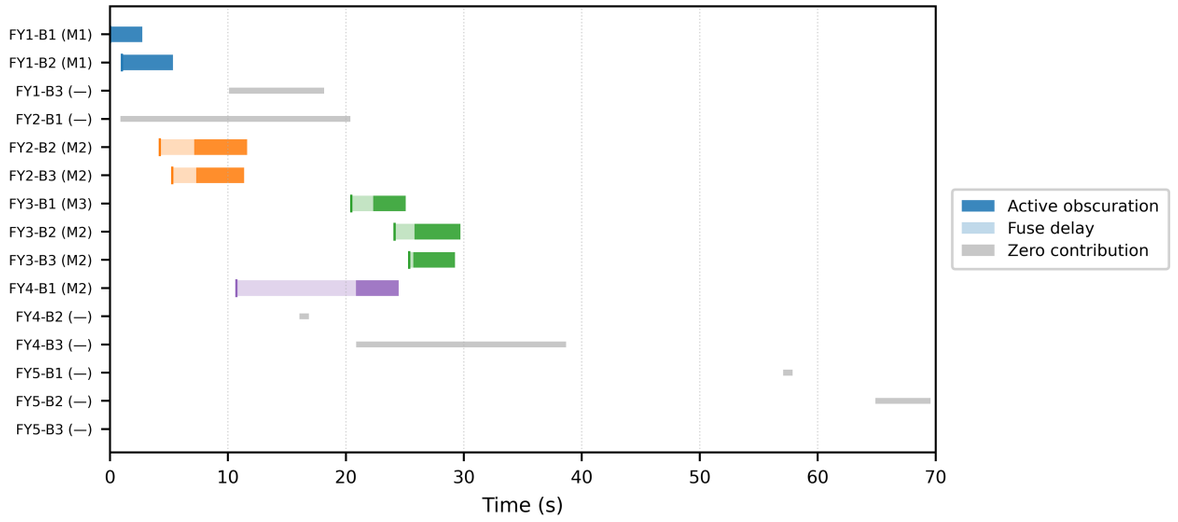
5. Discussion

5.1. *Emergent coordination and the logical-gate mechanism*

The strategy identified by our planner exhibits cooperative patterns that suggest a form of emergent coordination [57]. Rather than relying on static, greedy assignments, the UAV swarm displays a dynamic role allocation, shifting between distinct functional behaviors based on the evolving geometry of the engagement. Agents spontaneously adopted specialized roles, some engaging early threats to delay their advance while others established defensive screens to ensure continuous coverage, aligning with task specialization principles observed in heterogeneous multi-agent systems [58]. This



(a)



(b)

Fig. 6. Trajectory and control overview for the best-found 5-UAV cooperative strategy ($T_{\text{total}} = 28.73$ s). (a) 2D top-down engagement map. UAVs FY1–FY5 execute constant-heading, constant-speed open-loop commands (solid arrows; heading angle and speed listed in the legend); missiles M1–M3 approach the asset at 300 m/s along straight-line paths (dashed arrows). Drop points (×) and detonation points (•) are marked for all 15 munitions. (b) Deployment and obscuration timeline. Each bar spans from drop time (light fill) to end of the effective obscuration window (dark fill); the seven munitions with zero individual coverage contribution are shown in gray, and their cooperative role is analyzed in the Discussion (cf. Fig. 7).

division of labor appears to focus the collective effort of the system efficiently. Beyond this strategic allocation, a closer analysis of the solution reveals a less intuitive cooperative mechanism.

A closer analysis exposes the empirical signature of the nonsubmodular structure established in Sec. 3. Under the same optimization protocol, removing seven ostensibly nonproductive munitions and re-optimizing dropped the total obscuration from 28.73 to 22.25 s, even though each of these munitions has an individual effective obscuration time of zero. We refer to this phenomenon, in which an agent of zero standalone utility is nonetheless indispensable to the collective objective, as Phantom Synergy. The zero-coverage property of these munitions is measured directly from the optimized solution itself and is not assumed a priori. The comparison is also conservative, since the reduced-dimensional problem (26D) has a smaller search space than the full problem (40D), so under equal population size and generation budget each dimension receives more thorough exploration in the 26D case, yet the 40D solution consistently

Table 3. Mean objective evaluation time ± std (ms), reference Python implementation. $N_M = 3$, $N_t = 2000$, $N_p = 3$; 20 replicates per cell with random valid decision vectors.

N_U	$K = 1$	$K = 2$	$K = 3$
1	62 ± 1	124 ± 1	208 ± 15
2	146 ± 2	263 ± 14	373 ± 1
3	186 ± 1	373 ± 2	566 ± 2
4	251 ± 1	539 ± 33	756 ± 9
5	316 ± 5	628 ± 4	947 ± 38

outperforms it across runs. The effect is a direct consequence of the nondecomposable logical structure of the obscuration criterion. Unlike cumulative tasks where partial contributions add up, this problem functions as a logical gate.

Consider a formal scenario at time t involving a missile m, a primary munition A, a phantom munition B, and three critical asset points $\{p_1, p_2, p_3\}$. The individual effectiveness of munition A, denoted $I_{m,A}(t)$, is binary and requires all lines of sight to be intercepted:

$$I_{m,A}(t) = C_{m,A,p_1}(t) \wedge C_{m,A,p_2}(t) \wedge C_{m,A,p_3}(t). \quad (13)$$

A synergistic singularity occurs when munition A successfully intercepts two lines of sight (e.g. p_1, p_2) but fails the third (p_3), yielding $I_{m,A}(t) = 0$. At the same time, the phantom munition B is positioned to intercept only the complementary line of sight (p_3), also yielding a standalone utility of zero. The collective system effectiveness, $I_m(t)$, nonetheless evaluates the union of interceptions for each line of sight before assessing the conjunction:

$$I_m(t) = \bigwedge_{i=1}^{3} (C_{m,A,p_i}(t) \vee C_{m,B,p_i}(t)). \quad (14)$$

Substituting the conditions yields a successful collective outcome:

$$I_m(t) = (1 \vee 0) \wedge (1 \vee 0) \wedge (0 \vee 1) = 1 \wedge 1 \wedge 1 = 1. \quad (15)$$

This derivation, visually summarized in Fig. 7, illustrates that in high-dimensional logical control problems, the value of an agent is decoupled from its isolated output and is instead defined by its ability to complete a logical circuit within the swarm.

This principle is not unique to aerial defense. It mirrors phenomena in robotics, such as a robot applying zero net lifting force but providing essential stabilizing torque [59], or in sensor networks, where a sensor with no unique coverage is vital for bridging a logical gap between neighbors [60]. Phantom Synergy thus highlights the limitation of conventional, additive agent valuation methods. It underscores the challenge of credit assignment in multi-agent systems, where the marginal contribution of an agent is highly context-dependent, a difficulty well-documented in the reinforcement learning literature [61, 62].

SBOA-Pro serves as the solver, a metaheuristic optimizer suited to the nonconcave, black-box, high-dimensional objective. The contribution of this work is the problem formulation and the mechanism it reveals: the nondecomposable logical structure of the obscuration objective, in which an individual agent's utility is undefined in isolation. Because this structure is a property of the objective rather than of the solver, the Phantom Synergy effect is reproducible with any optimizer that reaches comparable solutions to the full 40D problem.

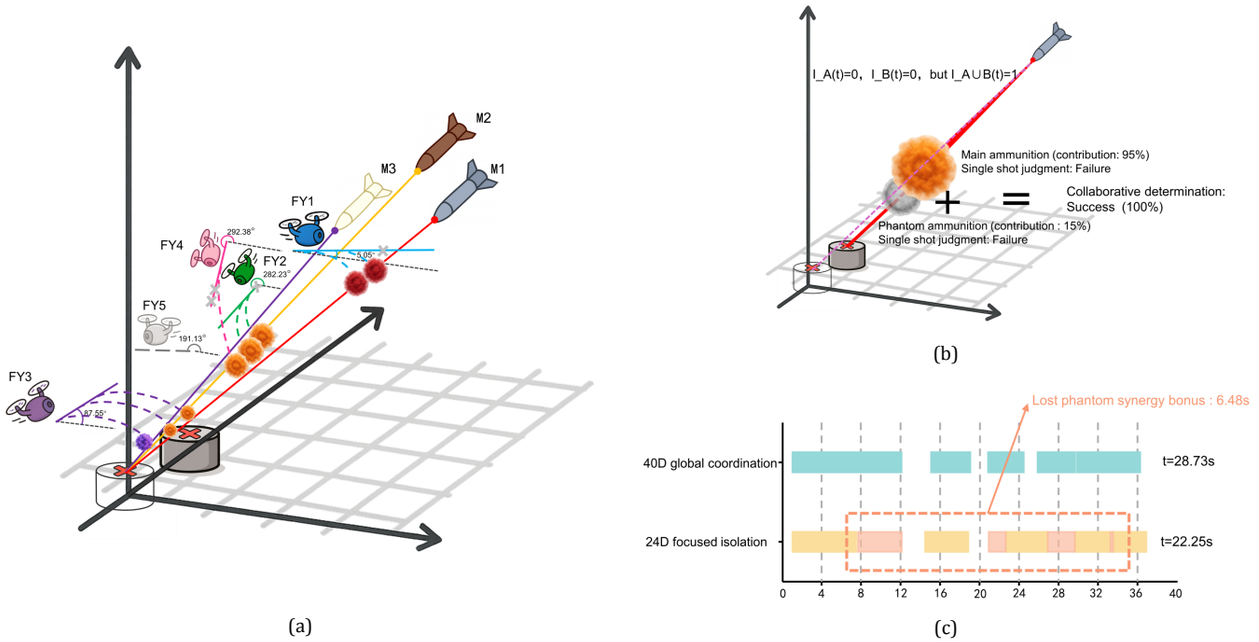
5.2. *Necessity of joint optimization*

The nonsubmodularity established in Proposition 3.4 predicts that agents cannot be valued in isolation, so the problem should not decompose into independently solvable sub-problems. The ablation results make this concrete. Removing the seven phantom munitions and re-optimizing (26D) yield only 22.25 s (a 22% loss), while the single-cloud analytical baseline gives $3 \times 5.36 \approx 16.1$ s (an independent-per-missile upper bound). The cooperative gain of joint optimization over independent single-cloud deployment thus exceeds 78%. A per-agent decomposition of the best-found strategy sharpens the picture. The five UAVs cover entirely distinct time windows, and evaluating each sub-vector independently yields individual contributions that sum exactly to the joint total of 28.73 s with zero temporal overlap. This perfect temporal partitioning was not imposed as a constraint but emerged from the joint optimization. Disjoint coverage windows do *not* imply that the problem is separable. Which window each UAV occupies is itself a product of the joint search, since the best window for any agent depends on what the others are already covering. The apparent independence of the final schedule is therefore a consequence of coordinated planning, not evidence that such planning was unnecessary.

5.3. *Post-hoc sensitivity analysis*

Because the strategy is open-loop, we characterize its tolerance to execution uncertainty via Monte Carlo perturbation of the decision vector (100 trials per configuration). Gaussian noise is applied simultaneously to all parameters: heading (σ_θ), speed (σ_v), and deployment/fuse timings (σ_t). The nominal performance is 28.73 s.

Table 4 reveals an asymmetry. Under UAV execution uncertainty (Mode A), the strategy degrades rapidly. Even modest heading noise ($\sigma_\theta = 0.5^\circ$) halves the nominal performance, because errors propagate linearly with predeployment flight time. A 0.5° heading error at $v = 127$ m/s over $t_{\text{drop}} = 20$ s produces a lateral displacement of ≈ 22 m, far exceeding $R_{\text{cloud}} = 10$ m. Per-agent analysis confirms that



(a) (b) (c)

Fig. 7. (Color online) The mechanism and impact of Phantom Synergy. (a) A visualization of the best-found strategy (28.73 s), where a phantom munition (gray trajectory) appears idle. (b) The micro-mechanism reveals that neither the primary munition (orange cloud) nor the phantom munition (gray cloud) can block all three lines of sight alone, but their spatial union creates a complete, synergistic blockade. (c) The Gantt chart from the counterfactual experiment quantifies this effect, showing the performance gap between the full 40-dimensional solution (cyan bar, 28.73 s) and the reduced 26-dimensional solution without phantom agents (yellow bar, 22.25 s).

Table 4. Sensitivity of the best-found strategy to execution uncertainty (Mode A) and threat uncertainty (Mode B). 100 Monte Carlo trials per row; nominal performance is 28.73 s.

Mode	σ_θ (deg)	σ_v (m/s)	σ_t (s)	Mean (s)	Std (s)	Min (s)	Retain
Nominal	0	0	0	28.73	0.00	28.73	100%
Mode A: UAV execution uncertainty (noise on decision vector)							
	0.5	1.0	0.1	13.00	4.07	5.91	45%
	1.0	3.0	0.3	7.74	2.64	3.85	27%
	2.0	5.0	0.5	5.94	2.40	0.00	21%
	3.0	7.0	0.7	4.30	2.27	0.00	15%
	5.0	10.0	1.0	3.85	2.43	0.00	13%
Mode B: Threat uncertainty ($V_M = 300 \pm \Delta v$)							
	$\Delta v = 5$ m/s			28.55	0.37	27.76	99%
	$\Delta v = 10$ m/s			28.35	0.68	26.59	99%
	$\Delta v = 20$ m/s			27.31	1.53	23.54	95%

early-deploying agents (FY1, $t_{\text{drop}} \approx 0$ s) are nearly insensitive, while late-deploying agents (FY3, $t_{\text{drop}} \approx 20$ s) dominate the degradation, and phantom agents (FY5, zero coverage) are unaffected because they contribute no coverage to lose. In contrast, threat uncertainty (Mode B) has negligible impact, since even a ± 20 m/s variation in missile speed (a 6.7% perturbation) retains 95% of nominal performance. This asymmetry indicates where the operational vulnerability lies. The strategy is robust to external threat variability but requires precise platform execution.

This study has several limitations. The model is deterministic and open-loop, so UAVs follow fixed heading and speed commands with no reactive adjustments, and the obscuration dynamics contain no stochastic elements. Missile trajectories are simplified to straight-line constant-velocity paths, whereas real intercept scenarios involve acceleration, maneuvers, and sensor-in-the-loop tracking. The model also omits environmental uncertainties such as wind-driven smoke dispersion [63], communication latencies, and sensor noise. No closed-loop stability or feasibility guarantees are provided, since the optimization yields open-loop plans without formal proof of robustness under perturbations or tracking error. The statistical validation is also limited by computational cost. The 40D problem was run two to five times under fixed computational budgets, insufficient to distinguish definitively between the full set of 15 munitions being necessary and the optimizer simply failing to find a competitive reduced-dimensional solution. The Phantom Synergy finding should therefore be read as evidence from the runs performed, not as statistical proof of necessity.

Future work should address these limitations along three directions. Robust and stochastic formulations are needed to account for environmental variability, following recent progress in distributed stochastic MPC for swarms [64] and chance-constrained obstacle avoidance [65]. Closed-loop receding-horizon control could replace the cur-

rent open-loop plan with adaptive policies that react to real-time sensor feedback and trajectory deviations. Multi-Agent Reinforcement Learning (MARL) offers a further path, training cooperative policies under perturbations by building on scalable methods [66–68] demonstrated in surveillance [69] and network positioning [70], so as to preserve Phantom Synergy benefits while mitigating brittleness.

6. Conclusion

We formulated cooperative multi-UAV obscuration as the maximization of a nonadditive union-of-occlusion objective, proved that its discrete selection layer is monotone but nonsubmodular, and solved the resulting 40-dimensional program with an enhanced metaheuristic, reaching a total obscuration of 28.73 s. Because the objective does not decompose, joint planning is necessary rather than optional, and an agent of zero individual utility can be essential to the collective outcome. This structural result is not specific to missile defense; it applies to any coverage task whose success is a conjunction of simultaneously required conditions, and it motivates solvers that exploit such structure instead of assuming additive utilities. Extending the open-loop plans to closed-loop receding-horizon control and to robust formulations under environmental uncertainty is the primary direction for future work.

Appendix A. SBOA-Pro Algorithm Details and Hyperparameters

To ensure the scientific rigor and reproducibility of our findings, this appendix provides a formal description of the SBOA-Pro algorithm and the specific hyperparameter configurations used in the full-scale (40D) optimization.

Appendix A.1. *Pseudocode for SBOA-Pro*

The SBOA-Pro algorithm (Algorithm A.1.) enhances the standard SBOA with two key mechanisms, namely elite seeding, handled during initialization and not shown in the main loop, and a flood diversity injection that counteracts stagnation. The pseudocode below details the main optimization loop, emphasizing the stagnation detection and response logic.

Algorithm A.1. Enhanced Secretary Bird Optimization Algorithm (SBOA-Pro)

Input: Objective function f, Dimension D, Population size N_{pop}, Search bounds [lb, ub]
Input Params: Max iterations $T_{\max}$, Stagnation threshold S_{thresh}, Elite retention ratio η
Output: Global best solution x_{best} and its fitness f_{best}

procedure SBOA-Pro-Optimize($f, D, N_{\text{pop}}, \ldots$)
  Initialize population X (size N_{pop}) randomly within [lb, ub]
  // Optional: Inject elite seeds from previous stages into X
  Evaluate initial fitness values $fit = f(X)$ in parallel
  Identify initial global best x_{best} and f_{best}
  Initialize stagnation counter $S_{\text{counter}} \leftarrow 0$
  Initialize last best fitness $f_{\text{last_best}} \leftarrow f_{\text{best}}$
  for $t = 1$ **to** $T_{\max}$ **do**
    Update population X_{new} using standard SBOA hunter and group dynamics
    Evaluate fitness $\text{fit}_{\text{new}} = f(X_{\text{new}})$ in parallel
    Update individual and global best solutions $(X, fit, x_{\text{best}}, f_{\text{best}})$
    // Stagnation detection and 'Flood' diversity injection
    if f_{best} has not improved compared to f_{last_best} **then**
      $S_{counter} \leftarrow S_{counter} + 1$
    else
      $f_{\text{last_best}} \leftarrow f_{\text{best}}$
      $S_{\text{counter}} \leftarrow 0$
    end if
    if $S_{\text{counter}} \geq S_{\text{thresh}}$ **then**
      Trigger Flood Mechanism:
      Preserve top $\eta \cdot N_{\text{pop}}$ elites first, then randomize the rest
      Identify and preserve top $\lfloor N_{\text{pop}} \cdot \eta \rfloor$ elite individuals
      Re-initialize the remaining $(N_{\text{pop}} - \lfloor N_{\text{pop}} \cdot \eta \rfloor)$ individuals randomly
      Re-evaluate the fitness of the new individuals in parallel
      Reset stagnation counter $S_{\text{counter}} \leftarrow 0$
    end if
  end for
  return $x_{\text{best}}, f_{\text{best}}$
end procedure

Table A.1. SBOA-Pro hyperparameter configuration.

Parameter	Symbol	Value	Description
Population Size	N_{pop}	10000	Total number of candidate solutions in the swarm.
Max Iterations	$T_{\max}$	50000	Maximum number of generations for evolution.
Stagnation Threshold	S_{thresh}	2000	Iterations without improvement to trigger the 'flood'.
Elite Retention Ratio	η	0.1	Proportion of elite solutions (10%) retained during a 'flood'.

Appendix A.2. *Hyperparameter configuration*

Table A.1 details the key hyperparameters for the SBOA-Pro algorithm used to solve the full-scale 40-dimensional optimization problem (Stage 4).

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Conflict of Interest

The authors have no relevant financial or nonfinancial interests to disclose.

Author Contributions

Yiquan Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing Original Draft, Writing Review & Editing.

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Data and Code Availability

The code is openly available on GitHub at https://github.com/wyqmath/cumcm2025.

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